**Type: DSC1-6(Major)**

**B.Sc(ECS)-II ( Semester IV )**

**Course Title: Relational Database Management System**

**(Paper Code: )**

**Credits: Theory – (2) Practical – (12)**

**Total Lectures:** 30 Hrs. **Contact Hrs. (L) :** 2

**University Evaluation:** 30 Marks **Internal Evaluation:** 20 Marks

**Course Outcomes:**

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| 1. | To Understand the PL/SQL architecture. |
| 2. | To learn programming, management, and security issues of working with PL/SQL program units, |
| 3. | To understand the built-in packages and user-defined packages. |
| 4. | To write PL/SQL code for procedures, triggers, cursors, exception handling, etc. |

| Unit | Content | Lect. |
|---|---|---|
| I | **Introduction of Transaction**:<br>ACID properties, transaction states, scheduling, and its types, conflict, and view serializability.<br>**Introduction of Concurrency Control:**<br>Problems of concurrency control, lock-based protocols, timestamp-based protocol, deadlock, and deadlock handling methods.<br>**Introduction, recovery algorithms**:<br>log base recovery, shadow paging, recovery with concurrent transactions, checkpoints or savepoints.<br>**Query Optimization:**<br>Overview Query Processing and Optimization – Heuristics and Cost Estimates in Query Optimization. | **15** |
| II | **Introduction to PL/SQL:**<br>Advantages, Architecture, Datatypes, Variable and Constants, Using Built_in Functions, Conditional, Looping and Iterations Statements, Selection Case, Simple Case, Goto Label and exit, SQL Within PL/SQL.<br>**Procedures in PL/SQL:** Stored Procedures, Procedure with Parameters (In, Out and In Out), Positional Notation and Named Notation, Dropping a Procedure.<br>**Functions in PL/SQL:** Difference between Procedures and Functions, types of functions and parameter modes,<br>**Packages in PL/SQL:** importance, advantages Implementing packages, Private and Public Objects in Package. | **15** |

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| | **Cursor in PL/SQL:** Types of Cursors, Cursor Attributes, Cursor with Parameters, Cursors with Loops Nested Cursors, Cursors with Sub Queries and Procedure.<br>**Exceptions in PL/SQL:** Types of exceptions, Raise_Application_Error, Pragma_Autonomous_Transaction<br>**Database Triggers in PL/SQL:** Types of Triggers, Row Level Triggers, Statement Level Triggers, Implementing triggers for various DML operations (insert, delete, update), DDL Triggers, Trigger Auditing. | |


| **Reference Books:** | |
|---|---|
| 1. | Database System Concepts by Korth Silberschetz |
| 2. | Fundamentals of Database Systems by Elmsari, Navathe |
| 3. | SQL, PL/SQL – The Programming Language of Oracle by Ivan Bayross |
| 4. | Database Management System by Seema Kedar |